

Centre of Gravity

- Centre of gravity: centre of gravity of a body is defined as the point through which resultant of the distributed gravitational forces act for any orientation of the body and denoted by C.G. The position of C.G depends upon the shape of body and it may occur inside or outside the body. A body has only one centre of gravity. Its location does not change even with a change in the orientation of the body.
- Centre of Mass: It is defined as the point where the whole mass of the body may be assumed to be concentrated.
- Centroid: The plane fig like rectangle, triangle circle etc have area only. The central point of the area of these plane figs. is known as centroid. In other words we can say the point where the entire area of the plane fig is assumed to be concentrated is called centroid.
- If the gravitational field is uniform and parallel for each part of the body and also the body has uniform density throughout then the C.G and C.M are coincident.
- If the thickness i.e mass of the body is not considered then the C.G and centroid pass through the same point

The centre of gravity and centroid may be found out by the following methods

1. By Method of Moments
2. By Method of Integration

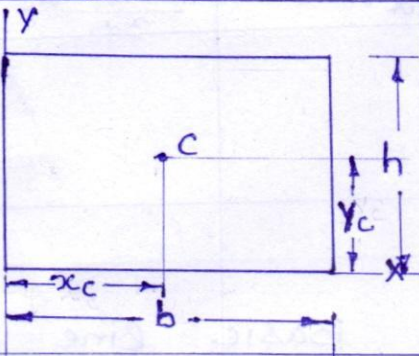
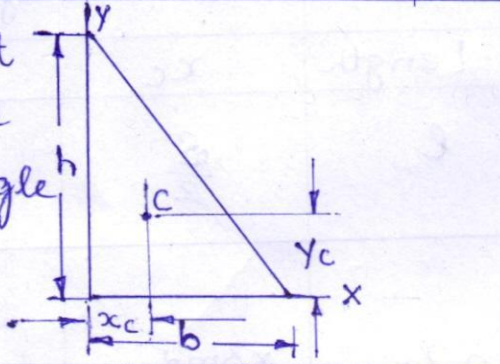
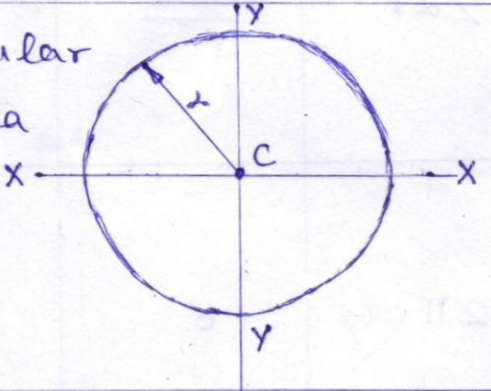
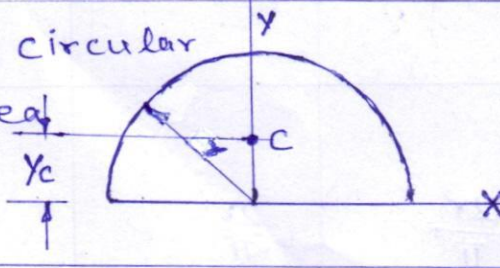
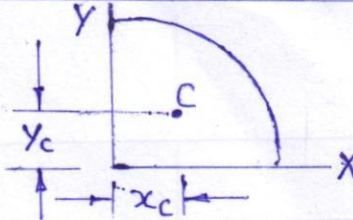
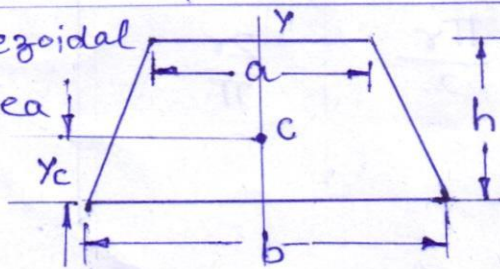
Generally C.G, Centroid, C.M are determined about two axis. These axis are called reference axis i.e the axis about which moment of area/mass are taken is known as reference axis.

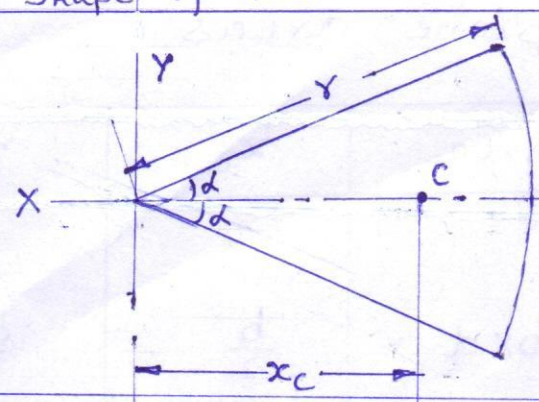
The axis of reference of a plane fig is generally taken as the lowest line of the fig for determining the y_c and left line of the fig for determining x_c .

x_c and y_c are the distances of centroid / C.G from the reference axis i.e. y -axis and x -axis respectively.

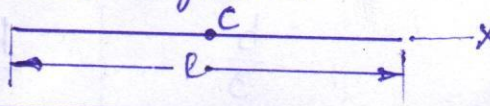
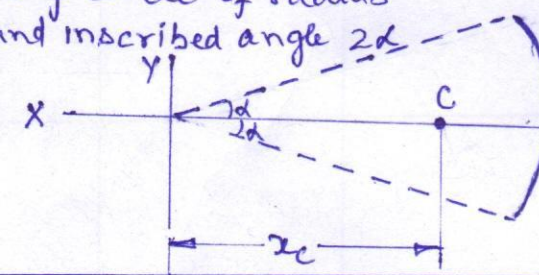
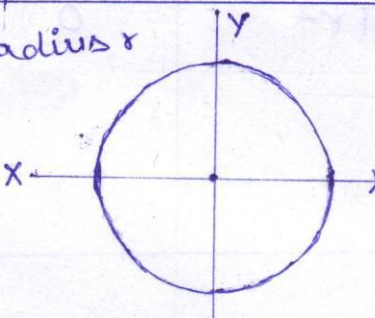
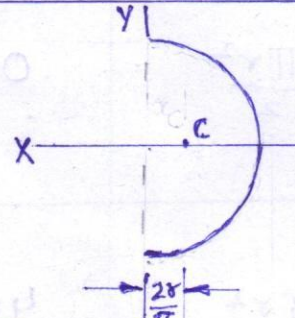
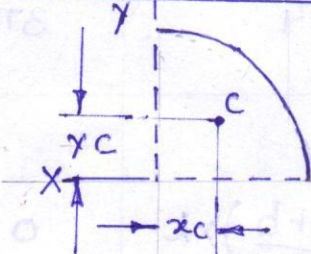
If the given fig is symmetrical about x -axis or y -axis then the C.G of the section will lie on the axis of symmetry.

Centroids of plane areas

S.No	Shape of area	Area	x_c	y_c
1.	Rectangular Area 	$b \times d$	$\frac{b}{2}$	$\frac{d}{2}$
2.	Right angle Triangle 	$\frac{bh}{2}$	$\frac{b}{3}$	$\frac{h}{3}$
3.	Circular Area 	πr^2	0	0
4.	Semi Circular Area 	$\frac{\pi r^2}{2}$	0	$\frac{4r}{3\pi}$
5.	Quarter circle 	$\frac{\pi r^2}{4}$	$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$
6.	Trapezoidal Area 	$\left(\frac{a+b}{2}\right) \times h$	0	$\frac{h}{3} \left(\frac{b+2a}{a+b} \right)$

	Shape of Area	Area	x_c	y_c
7		$\frac{2}{3} r^2 \alpha$	$\frac{2r \sin \alpha}{3\alpha}$	0

Centroid/C.G of Some Basic line Segments/Arcs

S.NO.	Description/Shape	Length	x_c	y_c
1	St line segment. 	l	$\frac{l}{2}$	0
2	Arc of circle of radius r and inscribed angle 2α . 	$2\alpha r$	$\frac{r \sin \alpha}{\alpha}$	0
3	Circle of Radius r . 	$2\pi r$	0	0
4		πr	$\frac{2r}{\pi}$	0
5		$\frac{\pi r}{2}$	$\frac{2r}{\pi}$	$\frac{2r}{\pi}$

Pappus and Guldinus Theorems

There are two theorems for determining

- (1) Surface area generated by a revolving curve about a non-intersecting axis
- (2) Volume generated by a revolving area about a non-intersecting axis.

1st theorem states that the area generated by revolving a plane curve about a non-intersecting axis in the plane of the curve is given by the product of length of the curve and the distance travelled by the centroid of the curve during revolution.

Mathematically we can write $A = H \times x_c \cdot \theta$

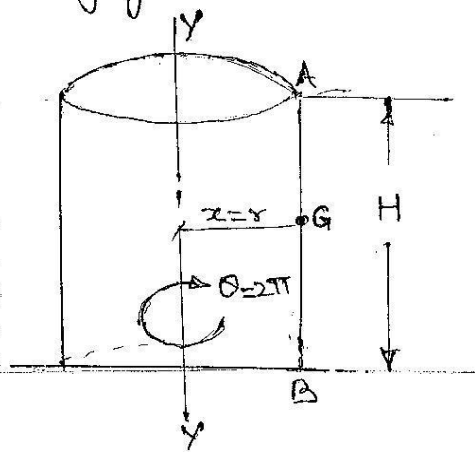
Where A is the Surface area generated

x_c is the distance of centroid of the curve from the axis

θ Angle of revolution in radians

To Explain the above theorem

Let a pt line AB of length H located at a distance $x=r$ from y -axis be rotated about y -axis for one complete revolution then we get a cylinder as shown in fig



Here Length $AB = H$

Distance of centroid of line AB from the y -axis is $x=r$

θ angle of revolution for a complete revolution of AB about y -axis

$$\theta = 2\pi$$

The area generated will be equal to

$$H \times r \times 2\pi = 2\pi r H$$

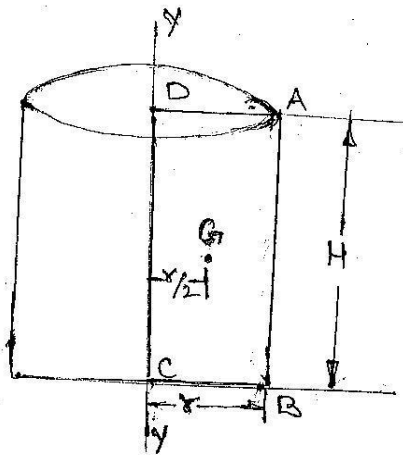
which is the surface area of a cylinder

Hence Proved.

2nd theorem states that the volume of a solid generated by revolving a plane area about a non-intersecting axis in the plane of area is given by the product of area of the plane fig and the distance travelled by the centroid of the area during revolution. Mathematically can write $V = A \cdot x_c \cdot \theta$.

Where V is the volume of the solid generated
 x_c is the distance of the centroid of the area from the axis
 θ angle of revolution

To explain the theorem let us assume a rectangular plane ABCD of height H and breadth r , located at a distance $x = r/2$ of the centroid of this area from the y -axis. Now let this area be rotated about y -axis for a complete revolution then we will get a solid known as cylinder as shown in fig



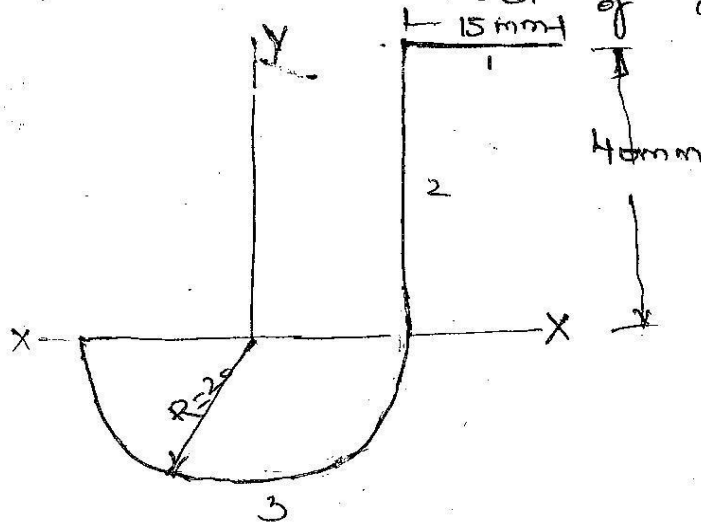
Now the area of plane ABCD $= H \times r$
the centroid of area from y -axis $= x = r/2$
and for a complete revolution of this area about y -axis $\theta = 2\pi$ radians
then the volume generated is

$$H \times r \times \frac{r}{2} \times 2\pi = \pi r^2 H$$

which is the volume of a solid cylinder

Hence the theorem is proved.

Determine the centroid of a rod shown in fig.



$$L_1 = 15 \text{ mm} = 15 \quad L_2 = 40$$

$$x_1 = 20 + 7.5 = 27.5 \quad x_2 = 20$$

$$y_1 = 40 = 40 \quad y_2 = 20$$

$$L_3 = \pi \times 20 = 62.83$$

$$x_3 = 0 = 0$$

$$y_3 = -\frac{2r}{\pi} = -\frac{40}{\pi} = -12.73$$

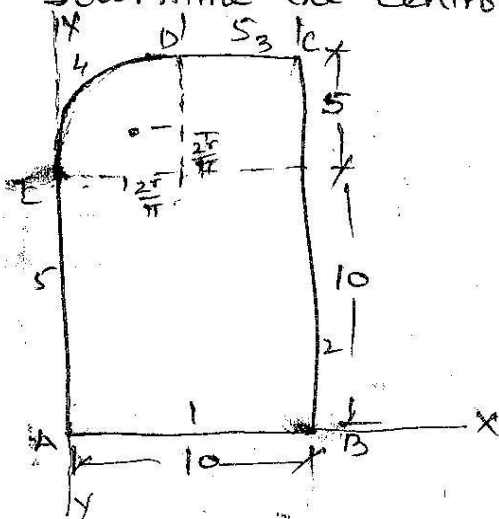
We know that

$$x_c = \frac{L_1 x_1 + L_2 x_2 + L_3 x_3}{L_1 + L_2 + L_3}$$

$$x_c = \frac{15 \times 27.5 + 40 \times 20 + 62.83 \times 0}{15 + 40 + 62.83} = 10.29 \text{ mm}$$

$$y_c = \frac{L_1 y_1 + L_2 y_2 + L_3 y_3}{L_1 + L_2 + L_3} = \frac{15 \times 40 + 40 \times 20 + 62.83 \times (-12.73)}{15 + 40 + 62.83} = \frac{600.17}{117.83} = 5.09 \text{ mm}$$

Determine the centroid of the wire as shown in fig



$$L_1 = 10 \text{ cm} \quad x_1 = 5 \text{ cm} \quad y_1 = 0$$

$$L_2 = 15 \text{ cm} \quad x_2 = 10 \quad y_2 = 7.5 \text{ cm}$$

$$L_3 = 5 \text{ cm} \quad x_3 = 7.5 \quad y_3 = 15 \text{ cm}$$

$$L_4 = \pi \times 5 = 7.85, \quad x_4 = 5 - \frac{2 \times 5}{\pi} = 1.816, \quad y_4 = 10 + \frac{2 \times 5}{\pi} = 13.183$$

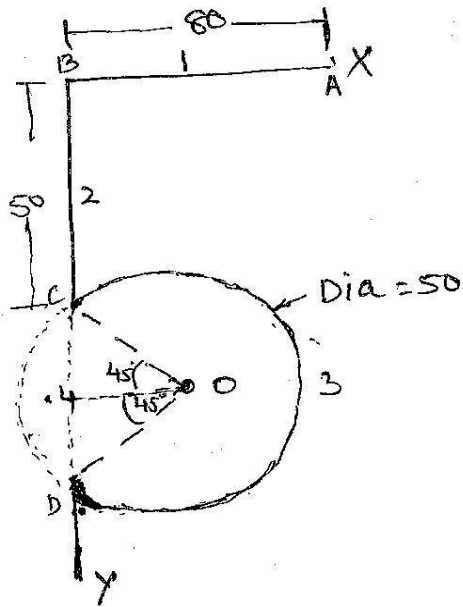
$$L_5 = 10 \quad x_5 = 0 \quad y_5 = 5 \text{ cm}$$

$$x_c = \frac{L_1 x_1 + L_2 x_2 + L_3 x_3 + L_4 x_4 + L_5 x_5}{L_1 + L_2 + L_3 + L_4 + L_5}$$

$$x_c = \frac{10 \times 5 + 15 \times 10 + 5 \times 7.5 + 7.85 \times 1.816 + 10 \times 0}{10 + 15 + 5 + 7.85 + 10} = \frac{251.75}{47.85} = 5.26$$

$$y_c = \frac{10 \times 0 + 15 \times 7.5 + 5 \times 15 + 7.85 \times 13.183 + 10 \times 5}{10 + 15 + 5 + 7.85 + 10} = \frac{340.98}{47.85} = 7.126 \text{ cm}$$

Determine the centroid of wire bent to form a number 5 with respect to the given axis. All dimensions are in mm



$$L_1 = 80 \quad x_1 = 40 \quad y_1 = 0$$

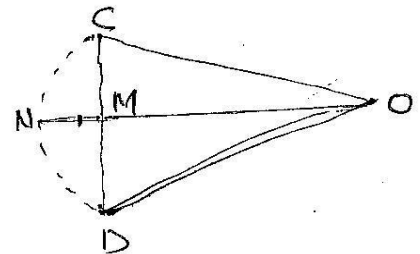
$$L_2 = 50 \quad x_2 = 0 \quad y_2 = -25$$

$$L_3 = 2\pi \times 25 = 314.159 \quad x_3 = r \cos 45 = 25 \cos 45 = 17.68 \quad y_3 = -r \sin 45 = -17.68$$

$$L_4 = \text{Arc CD} = 2\pi r = 2 \times 25 \times \pi/4 = 39.27$$

$$x_4 = -9.66$$

$$y_4 = -85.355$$



$$ON = \frac{r \sin 2\alpha}{2}$$

$$= \frac{50 \sin 45}{\pi/4} = 45.01$$

$$NM = 45.01 - 35.355 = 9.66$$

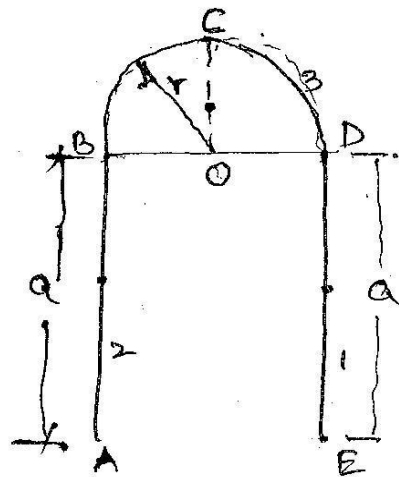
$$x_c = \frac{80 \times 40 + 50 \times 0 + 314.159 \times 17.68 - (39.27 \times -9.66)}{80 + 50 + 314.159 - 39.27}$$

$$= \frac{13548.49}{365.62} = 37.05$$

$$y_c = \frac{50 \times 0 + 50 \times -25 + 314.159 \times -17.68 - 39.27 \times -85.355}{365.62}$$

$$= \frac{-21360.40}{365.62} = -58.42 \text{ mm}$$

Determine the length of wire in terms of r such that the centroid of the fig show below is located at point O.



Taking moment of length about the line BD

$$L_1 = a \quad y_1 = -\frac{a}{2}$$

$$L_2 = a \quad y_2 = \frac{a}{2}$$

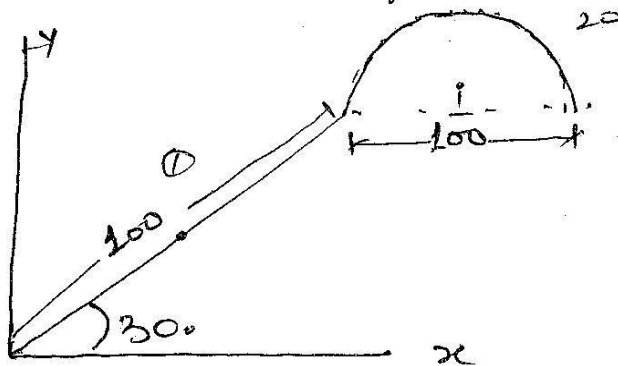
$$L_3 = \pi r \quad y_3 = \frac{2r}{\pi}$$

$$\Sigma M_{BD} = 0 \quad (\pi r) \times \left(\frac{2r}{\pi}\right) - (a \times \frac{a}{2}) \times 2 = 0$$

$$2r^2 = a^2 \Rightarrow a = \sqrt{2}r = \sqrt{2} \times r$$

$$\begin{aligned} \text{Total length of wire} &= a + a + \pi r = 2a + \pi r \\ &= 2\sqrt{2}r + \pi r = r(2\sqrt{2} + \pi) \end{aligned}$$

Find the centroid of a wire as shown in fig



$$L_1 = 100 \quad x_1 = 50 \cos 30 = 50 \times 0.866 = 43.3$$

$$y_1 = 50 \sin 30 = 50 \times 0.5 = 25$$

$$L_2 = \pi \times r = \pi \times 50 = 157$$

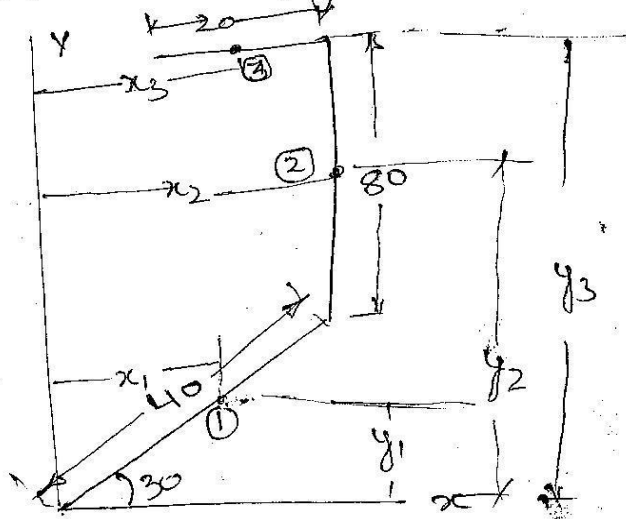
$$x_2 = 86.6 = 50 = 136.6$$

$$y_2 = 50 + 31.83 = 81.83$$

$$x_c = \frac{L_1 x_1 + L_2 x_2}{L_1 + L_2} = \frac{100 \times 43.3 + 157 \times 136.6}{100 + 157} = \frac{25746.2}{257} = 100.29$$

$$\begin{aligned} y_c &= \frac{L_1 y_1 + L_2 y_2}{L_1 + L_2} = \frac{100 \times 25 + 157 \times 81.83}{257} = \frac{15347.31}{257} \\ &= 59.71 \text{ cm} \end{aligned}$$

Determine the centroid of the wire as shown in fig. All dimensions are in cm.



$$x_1 = 20 \cos 30 = 17.32 \text{ cm}$$

$$y_1 = 20 \sin 30 = 10$$

$$x_2 = 40 \cos 30 = 34.64$$

$$y_2 = 40 \sin 30 + 40 = 60$$

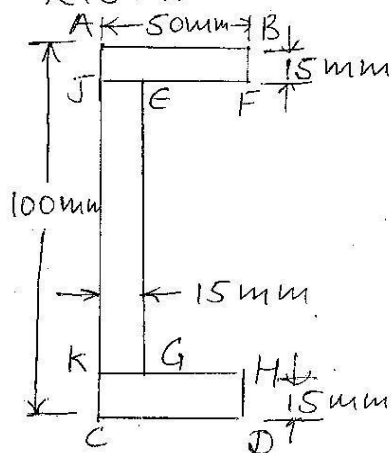
$$x_3 = 40 \cos 30 - 10 = 24.64$$

$$y_3 = 40 \sin 30 + 80 = 100$$

$$\text{Now } x_c = \frac{l_1 x_1 + l_2 x_2 + l_3 x_3}{l_1 + l_2 + l_3} = \frac{40 \times 17.32 + 80 \times 34.64 + 20 \times 24.64}{40 + 80 + 20} = 28.26 \text{ cm}$$

$$y_c = \frac{l_1 y_1 + l_2 y_2 + l_3 y_3}{l_1 + l_2 + l_3} = \frac{40 \times 10 + 80 \times 60 + 20 \times 100}{40 + 80 + 20} = 51.4 \text{ cm.}$$

Q-1. Find the centre of gravity of a channel section 100mm X 50mm X 15mm.



Solⁿ As the section is symmetrical about x-axis, therefore its centre of gravity will lie on this axis. Now split up the whole section into three rectangles ABFJ, EGTK and CDHK

Let the face AC be the axis of reference

(i) Rectangle ABFJ

$$a_1 = 50 \times 15 = 750 \text{ mm}^2$$

$$\text{and } x_1 = \frac{50}{2} \text{ mm}$$

(ii) Rectangle EGTK

$$a_2 = (100 - 30) \times 15 = 1050 \text{ mm}^2$$

$$\text{and } x_2 = \frac{15}{2} = 7.5 \text{ mm}$$

(iii) Rectangle CDHK :

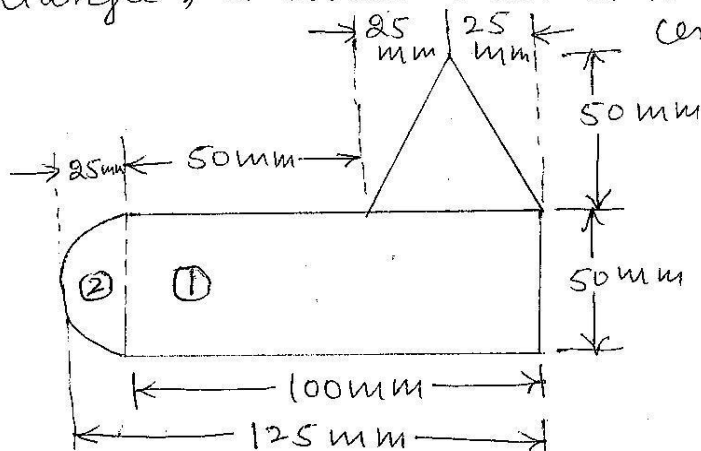
$$a_3 = 50 \times 15 = 750 \text{ mm}^2$$

$$x_3 = \frac{50}{2} = 25 \text{ mm}$$

We know that distance between the centre of gravity of the section and left face of the section AC.

$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3}{a_1 + a_2 + a_3} = \frac{750 \times 25 + (1050 \times 7.5) + (750 \times 25)}{750 + 1050 + 750} = 17.8 \text{ mm}$$

Q-2 A uniform lamina shown in given figure consist of a rectangle, a circle and a triangle. Determine the centre of gravity of lamina.



Solⁿ As the section is not symmetrical about any axis, therefore we have to find out the values of both $\bar{x}$ and $\bar{y}$ for the lamina.

Let left edge of circular portion and bottom face rectangular portion be the axes of references.

(i) Rectangular portion: +

$$a_1 = 100 \times 50 = 5000 \text{ mm}^2$$

$$x_1 = 25 + \frac{100}{2} = 75 \text{ mm}$$

$$\text{and } y_1 = \frac{50}{2} = 25 \text{ mm}$$

ii) Semicircular portion

$$a_2 = \frac{\pi}{2} \times r^2 = \frac{\pi}{2} (25)^2 = 982 \text{ mm}^2$$

$$x_2 = 25 - \frac{4r}{3\pi} = 25 - \frac{4 \times 25}{3\pi} = 14.4 \text{ mm}$$

$$\text{and } y_2 = \frac{50}{2} = 25 \text{ mm}$$

(iii) Triangular portion: +

$$a_3 = \frac{50 \times 50}{2} = 1250 \text{ mm}^2$$

$$x_3 = 25 + 50 + 25 = 100 \text{ mm}$$

$$y_3 = 50 + \frac{50}{3} = 66.7 \text{ mm}$$

We know that distance between centre of gravity of the section and left edge of the circular portion

$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3}{a_1 + a_2 + a_3} = \frac{(5000 \times 75) + (982 \times 14.4) + (1250 \times 100)}{5000 + 982 + 1250}$$

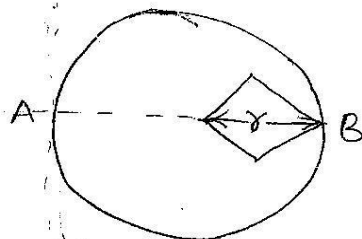
$$\therefore \bar{x} = 71.1 \text{ mm} \quad \underline{\text{Ans}}$$

Similarly $\therefore$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{a_1 + a_2 + a_3} = \frac{(5000 \times 25) + (982 \times 25) + (1250 \times 66.7)}{5000 + 982 + 1250}$$

$$= 32.2 \text{ mm} \quad \underline{\text{Ans}}$$

Q-3 A square hole is punched out of circular lamina the diagonal of the square being the radius of the circle as shown in given fig. Find the centre of gravity of the remainder, if r is the radius of the circle.



Soln As the section is symmetrical about x-axis. So its centre of gravity will lie on this axis. Let A be the point of reference.

i) Main circle

$$a_1 = \pi r^2$$

$$\text{and } x_1 = r$$

ii) cut out square

$$a_2 = \frac{r \times r}{2} = 0.5 r^2$$

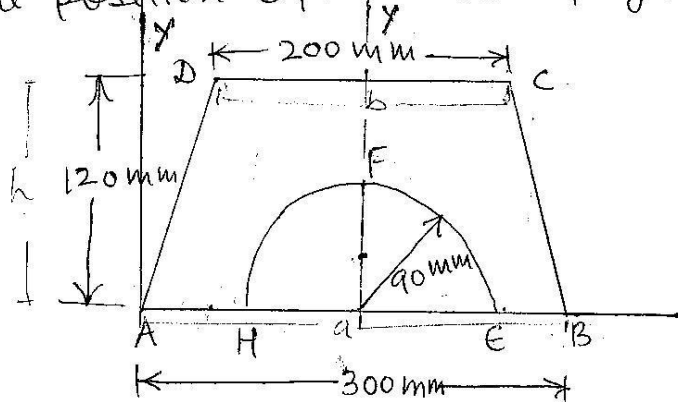
$$x_2 = r + \frac{r}{2} = 1.5 r$$

$$\text{So } \bar{x} = \frac{a_1 x_1 - a_2 x_2}{a_1 - a_2} = \frac{(\pi r^2 \times r) - (0.5 r^2 \times 1.5 r)}{\pi r^2 - 0.5 r^2}$$

$$= \frac{r^3 (\pi - 0.75)}{r^2 (\pi - 0.5)} = \frac{r (\pi - 0.75)}{\pi - 0.5} \quad \underline{\text{Ans}}$$

Q. 1 A Semicircle of 90mm radius is cut out from a trapezium as shown in fig:-

Find the position of centre of gravity.



$$a = h \times \left(\frac{a+b}{2} \right)$$

$$\bar{y} = \frac{h}{3} \left(\frac{a+2b}{a+b} \right)$$

As the section is symmetrical about Y-Y axis, therefore its centre of gravity will lie in this axis. Now consider two portions of fig, trapezium ABCD and semicircle EFH.

Let base of trapezium AB be the axis of reference.

(i) Trapezium ABCD

$$a_1 = \frac{120}{2} (200+300) = 30,000 \text{ mm}^2$$

$$\text{and } y_1 = \frac{120}{3} \times \left(\frac{300 + 2 \times 200}{300 + 200} \right) = 56 \text{ mm}$$

(ii) Semicircle EFH.

$$a_2 = \frac{1}{2} \times \pi r^2 = \frac{1}{2} \times \pi \times (90)^2 = 4050\pi \text{ mm}^2$$

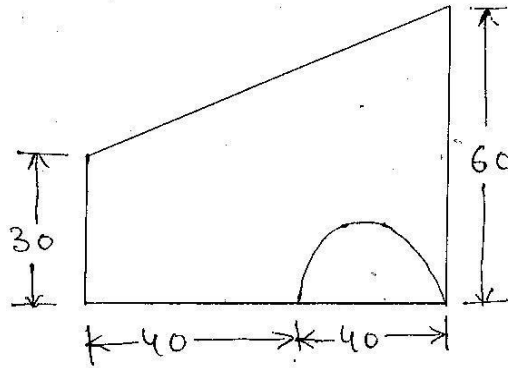
$$y_2 = \frac{48}{3\pi} = \frac{4 \times 90}{3\pi} = \frac{120}{\pi} \text{ mm}$$

We know that distance between centre of gravity of section and AB

$$\bar{y} = \frac{a_1 y_1 - a_2 y_2}{a_1 - a_2} = \frac{(30,000) 56 - (4050\pi \times \frac{120}{\pi})}{30,000 - 4050\pi} \text{ mm}$$

$$= 69.1 \text{ mm} \quad \underline{\text{Ans}}$$

Q-2 A semicircular area is removed from a trapezium as shown in fig. Determine the centroid of the remaining area. (Dimensions are in mm)



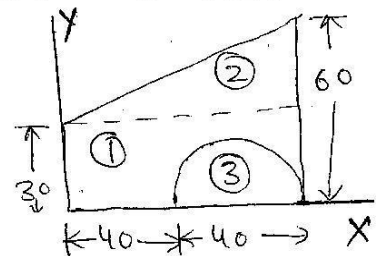
Solⁿ As the section is not symmetrical about any axis, therefore we have to find out the value of $\bar{x}$ and $\bar{y}$ for the area. Split up the area into three parts let left face and base of trapezium be the axes of reference.

(i) Rectangle

$$a_1 = 30 \times 80 = 2400 \text{ mm}^2$$

$$x_1 = \frac{80}{2} = 40 \text{ mm}$$

$$\text{and } y_1 = \frac{30}{2} = 15 \text{ mm}$$



(ii) Triangle

$$a_2 = \frac{1}{2} \times 80 \times 30 = 1200 \text{ mm}^2$$

$$x_2 = \frac{2}{3} \times 80 = 53.3 \text{ mm}$$

$$y_2 = 30 + \frac{30}{3} = 40 \text{ mm}$$

(iii) Semicircle:

$$a_3 = \frac{1}{2} \times \pi \times (20)^2 = 628.3 \text{ mm}^2$$

$$x_3 = 40 + \frac{40}{2} = 60 \text{ mm}$$

$$y_3 = \frac{48}{3\pi} = \frac{4 \times 20}{3\pi} = 8.5 \text{ mm}$$

point O.

Taking moments of the weights about line FA, we get

$$6W \times OH - W \times O = 5W \times GH$$

From the geometrical configuration,

$$OH = 1 \sin 60^\circ = 0.866 \text{ m and } AH = 0.5 \text{ m}$$

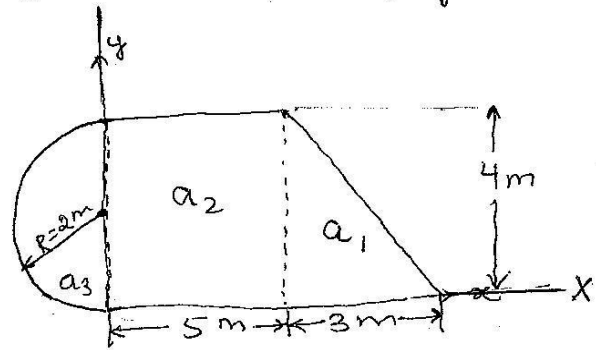
That gives $\rightarrow$

$$6W \times 0.866 = 5W \times GH$$

$$GH = \frac{6 \times 0.866}{5} = 1.039 \text{ m}$$

$$\begin{aligned} \therefore AG &= \sqrt{(AH)^2 + (GH)^2} \\ &= \sqrt{(0.5)^2 + (1.039)^2} \\ &= 1.15 \text{ m} \end{aligned}$$

Q) Locate the centroid of the area shown in fig with respect to the axes indicated in the fig.



Triangular Segment:

$$a_1 = \frac{1}{2} \times 3 \times 4 = 6 \text{ m}^2$$

$$x_1 = 5 + \frac{1}{3} \times 3 = 6 \text{ m}$$

$$y_1 = \frac{4}{3} = 1.33 \text{ m}$$

Rectangular Segment:

$$a_2 = 5 \times 4 = 20 \text{ m}^2$$

$$x_2 = \frac{5}{2} = 2.5 \text{ m}$$

$$y_2 = \frac{4}{2} = 2 \text{ m}$$

Semi-circular Segment:

$$a_3 = \frac{1}{2} \times \pi \times 2^2 = 6.28 \text{ m}^2$$

$$x_3 = -\frac{4R}{3\pi} = -\frac{4 \times 2}{3 \times \pi} = -0.849 \text{ m}$$

$$y_3 = \frac{4}{2} = 2 \text{ m}$$

The -ve with co-ordinate x_3 stems from the fact that this co-ordinate lies on the left of the y-axis.

$$\text{Then } \bar{x} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3}{a_1 + a_2 + a_3}$$

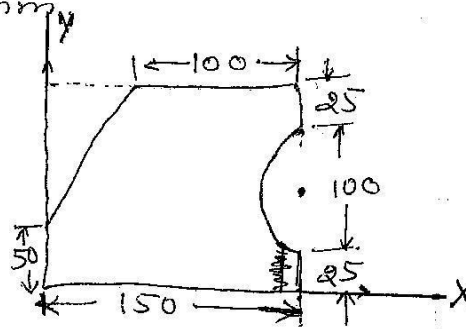
$$= \frac{6 \times 6 + 20 \times 2.5 + 6.28 (-0.849)}{6 + 20 + 6.28}$$

$$= 2.5 \text{ m}$$

$$\text{and } \bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{a_1 + a_2 + a_3}$$

$$= \frac{6 \times 1.33 + 20 \times 2 + 6.28 \times 2}{6 + 20 + 6.28} = 1.875 \text{ m}$$

Q) Locate the centroid of the area shown in fig. All the dimensions are in mm



Rectangle :

$$a_1 = 150 \times 150 = 22500 \text{ mm}^2$$

$$x_1 = \frac{150}{2} = 75 \text{ mm}$$

$$\text{and } y_1 = \frac{150}{2} = 75 \text{ mm}$$

$$\text{Semicircle : } a_2 = \frac{\pi r^2}{2} = \frac{\pi \times 50^2}{2} = 3925 \text{ mm}^2$$

$$x_2 = 150 - \frac{4r}{3\pi} = 150 - \frac{4 \times 50}{3\pi} = 128.77 \text{ mm}$$

$$y_2 = 25 + r = 25 + 50 = 75 \text{ mm}$$

$$\text{Triangle : } a_3 = \frac{1}{2}bh = \frac{1}{2} \times 50 \times 100 = 2500 \text{ mm}^2$$

$$x_3 = \frac{b}{3} = \frac{50}{3} = 16.67 \text{ mm}$$

$$y_3 = 150 - \frac{h}{3} = 150 - \frac{100}{3} = 116.67 \text{ mm}$$

$$\text{Then : } \bar{x} = \frac{a_1 x_1 - a_2 x_2 - a_3 x_3}{a_1 - a_2 - a_3}$$

$$= \frac{22500 \times 75 - 3925 \times 128.77 - 2500 \times 16.67}{22500 - 3925 - 2500}$$

$$\text{and } = 70.94 \text{ mm}$$

$$\bar{y} = \frac{a_1 y_1 - a_2 y_2 - a_3 y_3}{a_1 - a_2 - a_3}$$

$$= \frac{22500 \times 75 - 3925 \times 75 - 2500 \times 116.67}{22500 - 3925 - 2500}$$

$$= 68.52 \text{ mm}$$

Determine the co-ordinates of the centroid of the shaded area shown in fig

The given fig is the combination of a Δ and circular sector.

Area of ΔA_1

$$A_1 = \frac{1}{2} \times 30 \times 51.96 = 779.42$$

$$x_1 = 10, \quad y_1 = 51.96 \times \frac{2}{3} = 34.61$$

Area of circular sector

$$A_2 = \theta r^2$$

$$\frac{\pi \times 60^2}{360} \times 60 = 1884.95$$

$$\frac{\pi}{6} \times r^2 = \frac{\pi \times 60^2}{6} = 1884.95$$

x about O-1

$$= \frac{2r \sin \alpha}{3\alpha}$$

$$= \frac{2 \times 60 \times \sin 30}{3 \times \frac{\pi}{6}}$$

$$= \frac{2 \times 60 \times 0.5}{3 \times \pi} \times 6 = 38.19$$

$$x_2 = 38.19 \cos 30 = 33.07$$

$$y_2 = 38.19 \sin 30 = 19.09$$

$$\text{Now } x_c = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2}$$

$$= \frac{779.42 \times 10 + 1884.95 \times 33.07}{779.42 + 1884.95} = 26.32 \text{ cm}$$

$$y_c = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \frac{779.42 \times 34.61 + 1884.95 \times 19.09}{779.42 + 1884.95} = 23.60 \text{ cm}$$

Find the co-ordinates of centroid of the plane shaded area shown in fig

The given section is symmetrical about y-axis. Centroid lies on the axis of symmetry. Hence $x_c = 0$

Divide the given section in 4 parts

$$A_1 = \frac{1}{2} \times 15 \times 26 = 195 \text{ cm}^2$$

$$y_1 = \frac{26}{3} = 8.67 \text{ cm}$$

$$A_2 = 30 \times 26 = 780 \text{ cm}^2$$

$$y_2 = \frac{26}{2} = 13$$

$$A_3 = \frac{1}{2} \times 15 \times 26 = 195 \text{ cm}^2$$

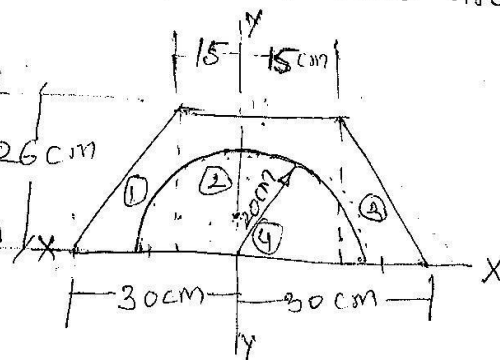
$$y_3 = \frac{26}{3} = 8.67$$

$$A_4 = \frac{\pi r^2}{2} = \frac{\pi \times 20^2}{2} = 628 \text{ cm}^2$$

$$y_4 = \frac{4R}{3\pi} = \frac{4 \times 20}{3\pi} = 8.49 \text{ cm}$$

Hence $x_c = 0$

$$y_c = 15.11 \text{ cm}$$



$$y_c = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3 - A_4 y_4}{A_1 + A_2 + A_3 + A_4}$$

$$= \frac{(195 \times 8.67) + (780 \times 13) + (195 \times 8.67) - (628 \times 8.49)}{195 + 780 + 195 - 628}$$

$$= \frac{1690.65 + 10140 + 1690.65 - 5331.72}{542} = 15.11$$

Find the co-ordinates of the centroid of the area left after removing a square area from a plate as shown in fig.

Considering the total plate

$$A_1 = a \times a = a^2$$

$$x_1 = \frac{a}{2}, y_1 = \frac{a}{2}$$

Removing Area

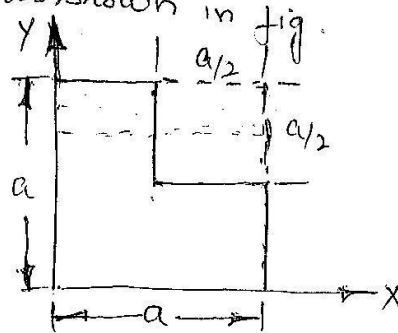
$$A_2 = \frac{a}{2} \times \frac{a}{2} = \frac{a^2}{4}$$

$$x_2 = \frac{a}{2} + \frac{a}{4} = \frac{3a}{4}$$

$$y_2 = \frac{a}{2} + \frac{a}{4} = \frac{3a}{4}$$

$$x_c = \frac{A_1 x_1 - A_2 x_2}{A_1 - A_2} = \frac{a^2 \times \frac{a}{2} - \frac{a^2}{4} \times \frac{3a}{4}}{a^2 - \frac{a^2}{4}} = \frac{\frac{a^3}{2} - \frac{3a^3}{16}}{\frac{3a^2}{4}} = \frac{\frac{8a^3 - 3a^3}{16}}{\frac{3a^2}{4}} = \frac{5a^3}{16} \times \frac{4}{3a^2} = \frac{5a}{12}$$

$$y_c = \frac{A_1 y_1 - A_2 y_2}{A_1 - A_2} = \frac{a^2 \times \frac{a}{2} - \frac{a^2}{4} \times \frac{3a}{4}}{a^2 - \frac{a^2}{4}} = \frac{\frac{a^3}{2} - \frac{3a^3}{16}}{\frac{3a^2}{4}} = \frac{5a^3}{16} \times \frac{4}{3a^2} = \frac{5a}{12}$$

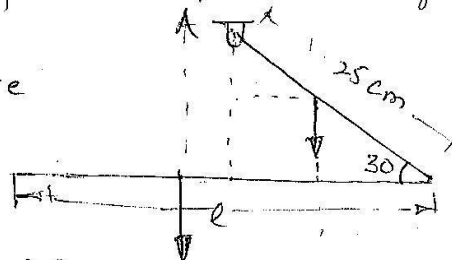


A homogeneous wire ABC is bent and attached to the hinge at A. Find the length l of the portion BC, for which portion BC of the wire is horizontal.

Let w be the wt per unit length of wire

Wt of wire 25cm long = $25w$

Wt of wire length $l = lw$



$$\text{Now } \sum M_A = 0 = 25w \times \frac{25}{2} \cos 30 = lw \times \left(\frac{l}{2} - 25 \cos 30 \right)$$

$$25w \times 10.82 = lw \left(\frac{l}{2} - 21.65 \right)$$

$$270.5 = \frac{l^2}{2} - 21.65l$$

$$\frac{l^2}{2} - 21.65l - 270.5 = 0$$

$$l^2 - 43.3l - 541 = 0$$

$$l = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= 43.3 \pm \sqrt{1874.89 + 4 \times 541}$$

$$\frac{43.3 + 63.55}{2} = 53.425$$

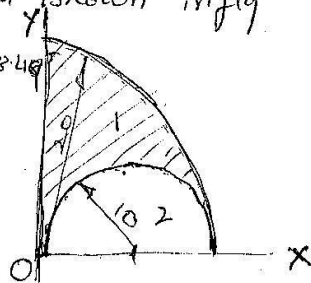
Find the coordinates of centroid of the shaded area shown in fig.

$$a_1 = \frac{\pi r^2}{4} = \frac{\pi \times 20^2}{4} = 314.16, \quad x_1 = \frac{4r}{3\pi} = \frac{4 \times 20}{3\pi} = 8.49, \quad y_1 = \frac{4r}{3\pi} = 8.49$$

$$a_2 = \frac{\pi r^2}{2} = \frac{\pi \times 10^2}{2} = 157, \quad x_2 = 10, \quad y_2 = \frac{4r}{3\pi} = \frac{40}{3\pi} = 4.24$$

$$x_c = \frac{a_1 x_1 - a_2 x_2}{a_1 - a_2} = \frac{314.16 \times 8.49 - (157 \times 10)}{314.16 - 157} = 6.98$$

$$y_c = \frac{a_1 y_1 - a_2 y_2}{a_1 - a_2} = \frac{314.16 \times 8.49 - 157 \times 4.24}{314.16 - 157} = 12.73$$

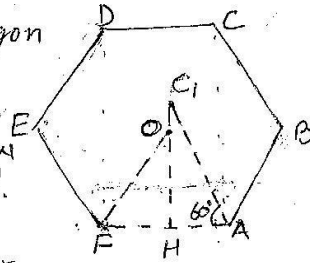


A rod of length 50 cm is bent so as to form the five sides of a regular hexagon ABCDEF as shown in fig. Find the centroid C_1 of it and determine the distance AC_1 .

A rod ABCDEF is bent to form a regular hexagon in which $AB = BC = CD = DE = EF = 10 \text{ cm}$.

Let w be the wt of each side. Hence total wt = $5w$ and acts at pt C_1 which represents the CG of rod.

Let a segment FA of 10 cm length and wt w be added and the regular hexagon be completed. For this hexagon the CG will be at pt O .



Now taking moment of the wts about line FA

M_{FA}

$$6W \times OH - W \times O = 5W \times C_1H$$

$$OH = 10/\sqrt{3} \sin 60 = 8.66$$

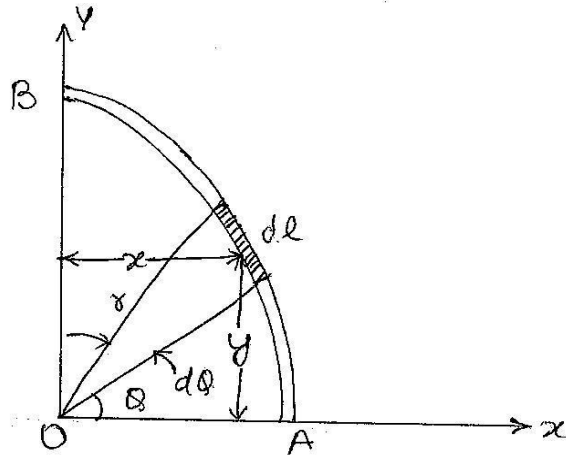
$$6W \times 8.66 = 5W \times C_1H$$

$$AH = 10/2 = 5 \text{ cm}$$

$$C_1H = \frac{6 \times 8.66}{5} = 10.39 \text{ cm}$$

$$\text{Now } AC_1 = \sqrt{(AH)^2 + (C_1H)^2} = \sqrt{5^2 + 10.39^2} = 11.53 \text{ cm}$$

Q- A uniform wire has been bent in the form of a quadrant of arc of a circle of radius r . Workout the location of its centroid.



Solⁿ The length AB of the wire can be divided into a number of small elements; one such element of length dl subtends at an angle dQ at O .

Centroidal distance of this element from x-axis

$$\bar{y} = \frac{\int y dl}{\int dl}$$

In terms of polar co-ordinates,

$$y = r \sin Q \quad \& \quad dl = r dQ$$

$$\text{then : } \int y dl = \int_0^{\pi/2} (r \sin Q) r dQ$$

$$= r^2 \int_0^{\pi/2} \sin Q dQ = r^2 [-\cos Q]_0^{\pi/2}$$

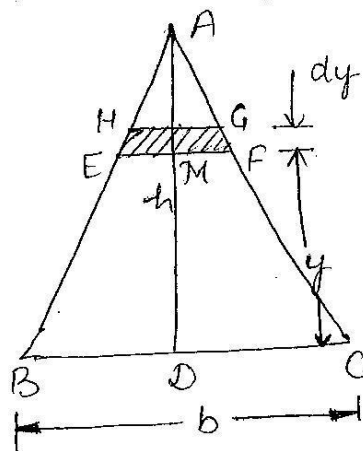
$$= r^2$$

$$\int dl = \int_0^{\pi/2} r dQ = r [Q]_0^{\pi/2} = r \frac{\pi}{2}$$

$$\therefore \bar{y} = \frac{r^2}{\frac{r\pi}{2}} = \frac{2r}{\pi}$$

Since the wire bent in the form of a quadrant of the arc of a circle is symmetrical about the x-axis & y-axis, $\bar{x} = \bar{y} = \frac{2r}{\pi}$

Centre of Gravity of a uniform triangular lamina



Considering a ΔABC whose base width is b and height is h . Consider an elementary strip EFGH of the lamina at a distance y from the base & of height dy .

Area of the elementary strip = $EF \times dy$

$$\text{Now } \frac{EF}{BC} = \frac{AM}{AD} \quad \text{or } EF = \frac{AM}{AD} \times BC$$

$$EF = \frac{b(h-y)}{h}$$

$$\therefore dA = \frac{b}{h} (h-y) dy$$

Let the Centre of Gravity of the lamina be at a distance $\bar{y}$ from the base. then

$$\begin{aligned} A\bar{y} &= \int y dA = \int_0^h \frac{b}{h} (h-y) y dy \\ &= \frac{b}{h} \int_0^h (hy - y^2) dy = \frac{b}{h} \left[\frac{hy^2}{2} - \frac{y^3}{3} \right]_0^h \end{aligned}$$

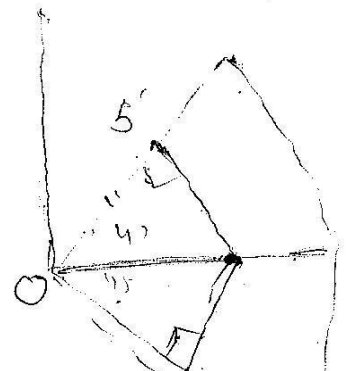
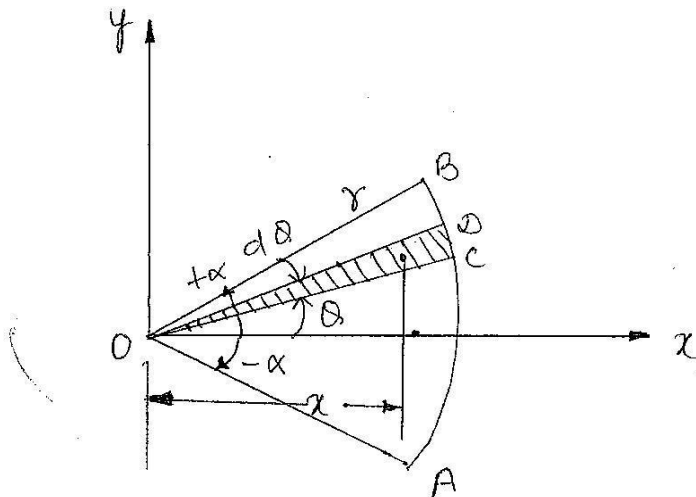
$$\Rightarrow A\bar{y} = \frac{b}{h} \left[\frac{h^3}{2} - \frac{h^3}{3} \right]$$

$$A\bar{y} = \frac{b}{h} \frac{h^3}{6} \Rightarrow \frac{b h^2}{2} \bar{y} = \frac{b h^2}{6} \times \frac{2}{bh}$$

$$\left(\because A = \frac{1}{2} \times b \times h \right)$$

$$\Rightarrow \bar{y} = \frac{h}{3}$$

Centroid of the Area of a sector of a circle having radius r and central angle 2α .



$$OQ' = \sqrt{2} \frac{4r}{3\pi} \cos 45^\circ = \frac{4r}{3\pi}$$

$$OB' = \sqrt{2} \frac{4r}{3\pi} \cos 45^\circ = \frac{4r}{3\pi}$$

Considering a sector OAB of a circle. Let OCD be an elementary strip of the sector between θ and $\theta + d\theta$.
Area of elementary strip

$$dA = \frac{1}{2} r \cdot CD = \frac{1}{2} \cdot r \cdot r d\theta = \frac{1}{2} r^2 d\theta$$

Centre of gravity of the elementary strip OCD, which is assumed triangular in shape is at a distance of $\frac{2}{3}r$ from O. Hence $x = \frac{2}{3} r \cos \theta$

Due to the symmetry of the sector about the x-axis, the centroid must lie on the line Ox. Let x_c be the distance of the centroid of the sector from O. Then

$$x_c = \frac{\int x dA}{\int dA} = \frac{\int_{-\alpha}^{+\alpha} \frac{2}{3} r \cos \theta \cdot \frac{1}{2} r^2 d\theta}{\int_{-\alpha}^{+\alpha} \frac{1}{2} r^2 d\theta} = \frac{\frac{2}{3} r \cdot \frac{1}{2} r^2 \int_{-\alpha}^{+\alpha} \cos \theta d\theta}{\frac{1}{2} r^2 \int_{-\alpha}^{+\alpha} d\theta}$$

$$\Rightarrow x_c = \frac{\frac{2}{3} r [\sin \theta]_{-\alpha}^{+\alpha}}{[\theta]_{-\alpha}^{+\alpha}} = \frac{\frac{2}{3} r [\sin \alpha - \sin(-\alpha)]}{[\alpha - (-\alpha)]}$$

$$= \frac{\frac{2}{3} r (2 \sin \alpha)}{2\alpha} = \frac{2}{3} r \frac{\sin \alpha}{\alpha}$$

For a semi-circle $\alpha = \frac{\pi}{2}$, then as shown on next page

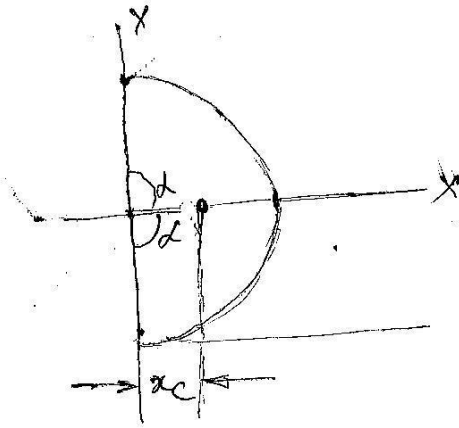
$$x_c = \frac{4r}{3\pi}$$

$$\alpha = \frac{\pi}{2}$$

$$x_c = \frac{2r \sin \alpha}{3\alpha}$$

$$= \frac{2r \sin 90}{3 \times \frac{\pi}{2}}$$

$$x_c = 2r \times \frac{2}{3\pi} = \frac{4r}{3\pi}$$



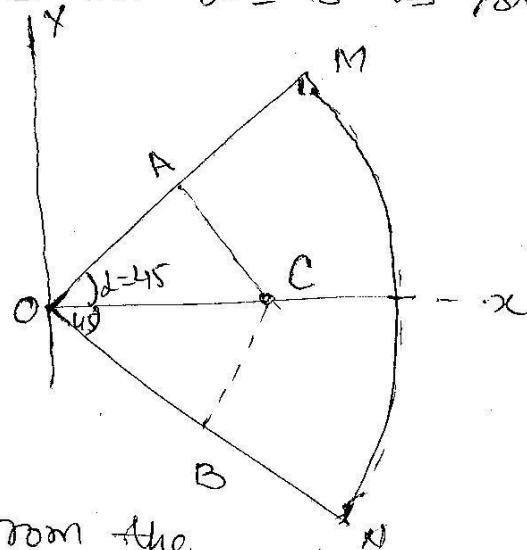
For the quadrant of a circle centroid can be determined as.

Centroid for any sector of the circle is $x_c = \frac{2r \sin \alpha}{3\alpha}$

For the quadrant of circle $\alpha = 45^\circ$ as shown in fig

Centroid of the quadrant from the origin O on the line O - x

$$= \frac{2r \sin 45}{3 \times \pi/4} = \sqrt{2} \times \frac{4r}{3\pi}$$



But the distance of centroid from the diametric axis - OM & ON i.e. distances OA & OB

$$OA = \sqrt{2} \times \frac{4r}{3\pi} \cos 45 = \sqrt{2} \times \frac{4r}{3\pi} \times \frac{1}{\sqrt{2}} = \frac{4r}{3\pi}$$

$$OB = \sqrt{2} \times \frac{4r}{3\pi} \sin 45 = \sqrt{2} \times \frac{4r}{3\pi} \times \frac{1}{\sqrt{2}} = \frac{4r}{3\pi}$$

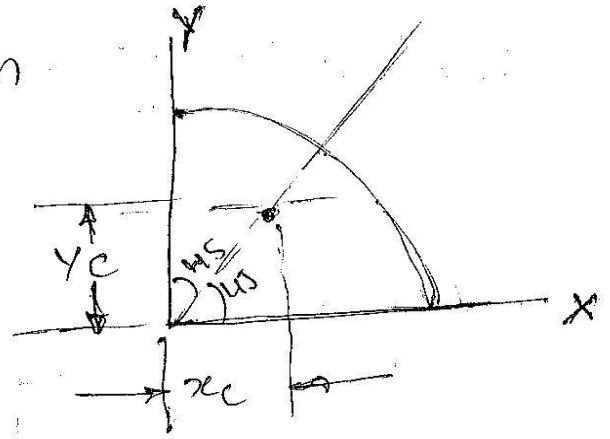
Hence

$$OA = \frac{4r}{3\pi}, OB = \frac{4r}{3\pi}$$

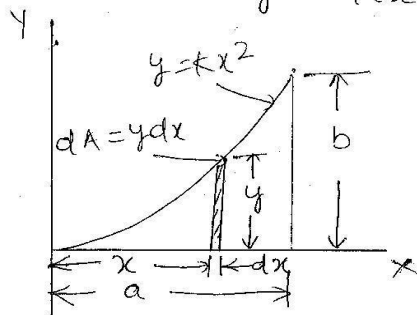
Distances of centroid from
the diametric axis

$$x_c = \frac{4r}{3\pi}$$

$$y_c = \frac{4r}{3\pi}$$



Q-1: Determine the centroid of the parabolic spandrel as shown in given fig. The eqn of the parabola is given by $y = kx^2$



Consider a vertical differential element of height y and width dx

Area of the strip $dA = y dx$

Distance of the centroid of the strip from the y -axis $= x$

$$x_c = \frac{\int x dA}{\int dA} = \frac{\int x \cdot y dx}{\int y dx} = \frac{\int_0^a x \left(\frac{b}{a^2} x^2 \right) dx}{\int_0^a \frac{b}{a^2} x^2 dx}$$

$$x_c = \frac{\left[\frac{b}{a^2} \frac{x^4}{4} \right]_0^a}{\left[\frac{b}{a^2} \frac{x^3}{3} \right]_0^a}$$

$$x_c = \frac{3}{4} a \text{ Ans}$$

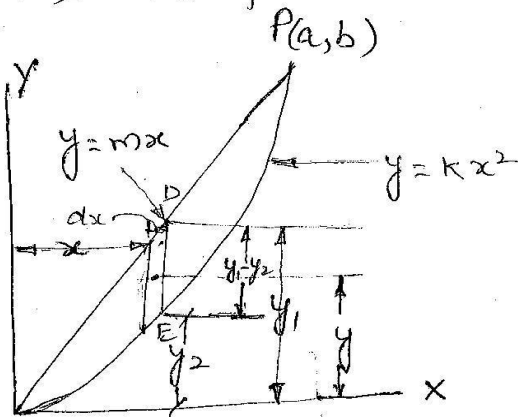
$$y_c = \frac{\int y dA}{\int dA}$$

Distance of the centroid of the strip from the x -axis $= y/2$

$$y_c = \frac{\int y/2 dx}{\int y dx} = \frac{\int_0^a \frac{1}{2} y^2 dx}{\int_0^a y dx} = \frac{\int_0^a \frac{1}{2} \left(\frac{b}{a^2} x^2 \right)^2 dx}{\int_0^a \frac{b}{a^2} x^2 dx}$$

$$y_c = \frac{\left[\frac{b^2}{2a^4} \frac{x^5}{5} \right]_0^a}{\left[\frac{b}{a^2} \frac{x^3}{3} \right]_0^a} \Rightarrow y_c = \frac{3}{10} b \text{ Ans}$$

Determine the co-ordinates of centroid of the shaded area formed by the intersection of a st line and the curve as shown in fig by method of Integration. If eqn of st line is $y = mx$ and for the curve $y = Kx^2$



Now we know that the coordinates of centroid of the given shaded area can be determined as

$$x_c = \frac{\int x dA}{\int dA}$$

$$y_c = \frac{\int y dA}{\int dA}$$

The area of Strip = $(y_1 - y_2) \times dx$

and the distance of centroid of Strip from y axis is x

$$\therefore x_c = \frac{\int_0^a x (y_1 - y_2) dx}{\int_0^a (y_1 - y_2) dx} = \frac{\int_0^a \left(\frac{b}{a} x - \frac{b}{a^2} x^2 \right) x \cdot dx}{\int_0^a \left(\frac{b}{a} x - \frac{b}{a^2} x^2 \right) dx} = \frac{\int_0^a \left(\frac{b}{a} x^2 - \frac{b}{a^2} x^3 \right) dx}{\int_0^a \left(\frac{b}{a} x - \frac{b}{a^2} x^2 \right) dx}$$

$$x_c = \frac{\left[\frac{b}{a} \cdot \frac{x^3}{3} - \frac{b}{a^2} \cdot \frac{x^4}{4} \right]_0^a}{\left[\frac{b}{a} \cdot \frac{x^2}{2} - \frac{b}{a^2} \cdot \frac{x^3}{3} \right]_0^a} = \frac{\left[\frac{b}{a} \cdot \frac{a^3}{3} - \frac{b}{a^2} \cdot \frac{a^4}{4} \right]}{\left[\frac{b}{a} \cdot \frac{a^2}{2} - \frac{b}{a^2} \cdot \frac{a^3}{3} \right]} = \frac{\frac{ba^2}{3} - \frac{ba^2}{4}}{\frac{ba}{2} - \frac{ba}{3}}$$

$$x_c = \frac{\frac{ba^2}{12}}{\frac{ba}{6}} = \frac{ba^2}{12} \times \frac{6}{ba} = \boxed{\frac{a}{2}}$$

The Point P lies on both the curves.

Hence For eqn of st line

$$y = mx$$

at $P(a, b)$

$$b = ma$$

$$m = \frac{b}{a}$$

Take a thin vertical strip as shown in fig

Point D lies on straight line

$$\therefore y_1 = mx \quad \text{or} \quad \boxed{y_1 = \frac{b}{a} x}$$

Point E lies on the curve or parabola having equation

$$y = Kx^2 \quad \text{at } P(a, b)$$

$$b = Ka^2 \quad \text{or} \quad K = \frac{b}{a^2}$$

$$\text{Hence} \quad \boxed{y_2 = \frac{b}{a^2} x^2}$$

Now the distance of centroid of the strip from X axis is y

$$y = y_2 + \frac{y_1 - y_2}{2} = \frac{2y_2 + y_1 - y_2}{2} = \frac{y_1 + y_2}{2}$$

$$y_c = \frac{\int_0^a \frac{y_1 + y_2}{2} x \, dA}{\int_0^a dA} = \frac{\frac{1}{2} \left(\frac{b}{a} x + \frac{b}{a^2} x^2 \right) \left(\frac{b}{a} x - \frac{b}{a^2} x^2 \right) dx}{\left(\frac{b}{a} x - \frac{b}{a^2} x^2 \right) dx}$$

$$y_c = \frac{\int_0^a \frac{1}{2} \left(\frac{b^2}{a^2} x^2 - \frac{b^2}{a^4} x^4 \right) dx}{\int_0^a \left(\frac{b}{a} x - \frac{b}{a^2} x^2 \right) dx} \quad \left(\because (a+b)(a-b) = a^2 - b^2 \right)$$

$$y_c = \frac{\frac{1}{2} \left[\frac{b^2}{a^2} \frac{x^3}{3} - \frac{b^2}{a^4} \frac{x^5}{5} \right]_0^a}{\left[\frac{b}{a} \frac{x^2}{2} - \frac{b}{a^2} \frac{x^3}{3} \right]_0^a} = \frac{\frac{1}{2} \left[\frac{b^2}{a^2} \cdot \frac{a^3}{3} - \frac{b^2}{a^4} \cdot \frac{a^5}{5} \right]}{\left[\frac{b}{a} \cdot \frac{a^2}{2} - \frac{b}{a^2} \cdot \frac{a^3}{3} \right]}$$

$$= \frac{\frac{1}{2} \left(\frac{b^2 a}{3} - \frac{b^2 a}{5} \right)}{\frac{ba}{2} - \frac{ba}{3}} = \frac{\frac{1}{2} \left(\frac{2b^2 a}{15} \right)}{\frac{ba}{6}}$$

$$y_c = \frac{b^2 a}{15} \times \frac{6}{ba} = \boxed{\frac{2b}{5}}$$